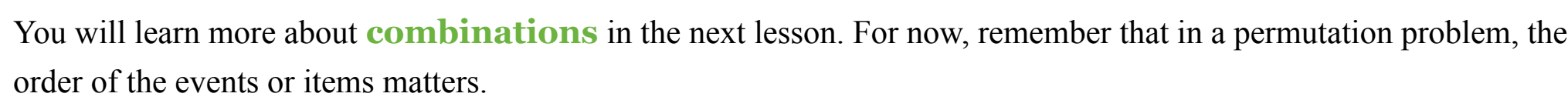


In real life, the word *combination* is often used when really it is a *permutation*. When you use a combination lock, you have to enter the numbers in a certain order. A combination lock should really be called a permutation lock.



There are 120 different ways the books can be arranged on the shelf.

To solve any permutation problem, simply multiply the choices. Study how this basic idea can be used to solve a more complex problem.

There are **2,730 different ways** the runners could finish 1st, 2nd, and 3rd.

Will the choices always count down? Not always.

There are 10 choices for each dial: $10 \times 10 \times 10 = \mathbf{1,000}$.

